

AI Assignment #2

Variational Bayesian Last Layers for Clothing Image Classification

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CISC3024 Pattern Recognition

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Abstract

Thesis: a discriminative variational Bayesian last layer (D-VBLL; Harrison, Willes, and Snoek, ICLR 2024) puts a Gaussian posterior on the last-layer weights of a clothing classifier. On Fashion-MNIST it matches softmax MAP on accuracy, likelihood, and on which errors it ranks as uncertain. Its calibration error is lower than an unscaled softmax, but one temperature fit on the validation set calibrates the softmax further. Gaussian naive Bayes on pixels fails; linear discriminant analysis on the same CNN features is close in accuracy and worse as a probability model. The learned weight variance stays far below the prior, so that discriminative posterior stays close to a point estimate. The generative VBLL from the same paper, which applies Bayes' rule to class-conditional Gaussians on the features, is the head that separates MNIST from clothes. Neither head detects low-contrast shift, and refusing low-confidence predictions does not repair rotation. Search, implementation, experiments, and this report were done by a Cursor agent under a standing instruction that I would not hand-write code.

1 How I Asked AI Tools to Find the Algorithm

The Moodle brief asks for a *recent* classification or recognition algorithm based on probability or Bayes theory, for computer vision, with the search, code, experiments, and report done by an AI tool. I used Cursor only. I pasted the brief, pointed at my Assignment 1 repository, and gave the agent authority to choose the method. Assignment 1 had already used FasterNet, a CVPR 2023 convolution designed for speed. That model is not a probabilistic classifier, so it could not be reused as the answer to this brief.

The agent searched recent Bayesian deep-learning papers and kept three candidates that are actually about classification uncertainty:

- **Variational Bayesian Last Layers** (Harrison, Willes, and Snoek, ICLR 2024), a sampling-free variational posterior on the last layer.
- **FSP-Laplace** (Cinquin et al., NeurIPS 2024), a Laplace approximation with a function-space Gaussian-process prior.
- **Last Layer Empirical Bayes** (Villego et al., 2025), a normalizing-flow prior on the last layer.

The choice was a table, not a preference for the newest title. D-VBLL is the only one of the three whose training loss is a closed form I can check on a toy batch before spending a training run. FSP-Laplace needs a function-space mode and a matrix-free Hessian. Last-layer empirical Bayes adds a normalizing flow on top of the classifier. Fashion-MNIST clothing recognition is also a different task from the digit-robustness study in Assignment 1.

Table 1: Candidates considered for a recent Bayes classifier that a laptop run can verify.

	D-VBLL	FSP-Laplace	LLEB
Venue	ICLR 2024	NeurIPS 2024	2025
Object with a posterior	last-layer W	network function	last-layer flow
Check before training	Theorem 2	weak mode, Hessian	ELBO through a flow
Extra machinery	diagonal Gaussian	function-space GP	normalizing flow
Used in this report	yes	no	no

The diary of prompts, rejected alternatives, and checks is `process_log.md`.

2 Algorithm Description

Model. A network with weights θ maps an image x to a feature vector $\phi(x, \theta) \in \mathbb{R}^D$. Class logits are a noisy linear readout. For a last-layer matrix W whose k -th row is w_k ,

$$p(y \mid x, W, \theta) = \text{softmax}(z), \quad z = W\phi(x, \theta) + \varepsilon, \quad (1)$$

with ε zero-mean Gaussian noise. This is Bayesian logistic regression on learned features. The prior on each row is isotropic Gaussian, $p(w_k) = \mathcal{N}(0, s^2 I)$, with the authors’ fan-in scaling $s^2 = 2/D$. Exact softmax marginalization is intractable, so training uses a variational posterior that factorizes across classes, $q(w_k) = \mathcal{N}(m_k, \text{diag}(v_k))$. Diagonal covariances drop cross-class correlation and make every loss term linear in D .

Training bound. Stochastic variational inference [7] maximizes a lower bound on the log marginal likelihood. For one example with label y , Theorem 2 replaces the expected log-softmax by

$$m_y^\top \phi - \log \sum_k \exp\left(m_k^\top \phi + \frac{1}{2}(\phi^\top \text{diag}(v_k)\phi + \sigma_k^2)\right). \quad (2)$$

The extra $\frac{1}{2}\phi^\top \text{diag}(v_k)\phi$ term is the Jensen gap: uncertain weights inflate every logit before the log-sum-exp, which stops the bound from behaving like an over-confident point estimate. The mini-batch loss I minimize is the negative of that bound, plus $(1/T) \text{KL}(q(W) \parallel p(W))$, minus a $(1/T)$ inverse-Wishart-style penalty on the noise variances σ_k^2 . T is the number of training images. Feature weights θ get an ordinary Gaussian prior, implemented as AdamW weight decay. The last-layer parameters are not weight-decayed, because the KL term is already their regularizer. This is the paper’s “full training” objective, not a post-hoc Laplace approximation and not Bayes-by-Backprop on every layer [6].

Prediction. The posterior predictive probability is the expectation of the softmax under $q(W)$. That expectation has no closed form for the discriminative model, so I draw logit samples

$$z^{(s)} \sim \mathcal{N}(m^\top \phi, \text{diag}(\phi^\top S \phi + \sigma^2)) \quad (3)$$

and average $\text{softmax}(z^{(s)})$. Test metrics use 32 samples. Out-of-distribution scores use the maximum predictive probability [4].

What is Bayesian here. Write the four objects separately, because a pattern-recognition classifier is the posterior predictive, not the training loss. The prior is $p(W) = \prod_k \mathcal{N}(w_k; 0, s^2 I)$. The likelihood is the softmax in Equation (1). The variational posterior $q(W)$ stands in for $p(W \mid \text{data})$, and the KL term keeps it from ignoring the prior. The reported class probability is the posterior predictive $\mathbb{E}_{q(W)}[\text{softmax}(W\phi)]$, not a point W . A softmax MAP baseline uses the same features and the same likelihood, but one weight matrix. Gaussian naive Bayes and linear discriminant analysis, used later as controls, are the generative version of this story: they put Gaussians on ϕ or on the

pixels and apply Bayes' rule for $p(y | x)$. D-VBLL is the discriminative version. It never builds $p(x | y)$.

Generative VBLL. The same paper also specifies a generative classifier. Class k owns a Gaussian mean μ_k in feature space, with prior $\mathcal{N}(0, s^2 I)$ and variational posterior $\mathcal{N}(\bar{\mu}_k, \text{diag}(v_k))$. A shared diagonal noise covariance Σ gives the marginal $p(\phi | k) = \mathcal{N}(\bar{\mu}_k, \Sigma + \text{diag}(v_k))$. With a uniform class prior, which is exact on this balanced training split, Bayes' rule is a softmax of those log densities:

$$p(y | \phi) = \text{softmax}_y(\log p(\phi | y)). \quad (4)$$

Theorem 3 trains this model with a sampling-free bound. The predictive distribution is closed form, so there is no Monte Carlo step. I call this head G-VBLL. It is the deep analogue of the LDA control.

Discriminative VBLL classifier used in this assignment

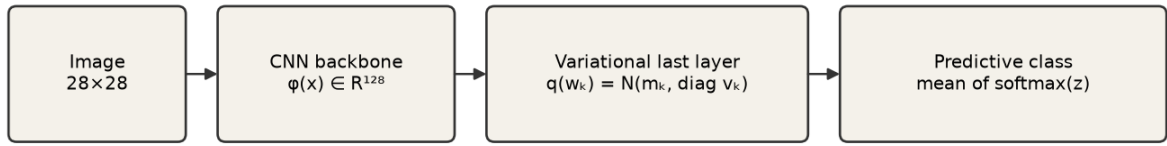


Figure 1: Student-scale D-VBLL. The MAP baseline replaces the variational last layer with one linear layer and a softmax.

3 How AI Implemented the Algorithm

The Cursor agent wrote the repository. I did not hand-write or hand-edit the Python. The layout is:

- `models/vbll.py` — diagonal D-VBLL (Theorem 2, Monte Carlo softmax) and G-VBLL (Theorem 3, closed-form Bayes rule).
- `models/backbone.py` — shared CNN. Three conv stages, batch-norm, global average pool, 128-d features. About 1.58×10^5 parameters.
- `models/classifier.py` — `map`, `vbll`, or `gvbll`. Weight decay applies to the backbone only when the head is Bayesian.
- `data_utils.py` — Fashion-MNIST / MNIST tensors, shared augmentation, pixel-space corruptions.
- `train.py`, `evaluate.py`, `metrics.py`, `figures.py`.
- `run_all.py`, `run_ablation.py`, `run_upgrades.py`.
- `extra_analysis.py`, `upgrade_analysis.py`. Neither script retrains a finished checkpoint.

The training step, in the form that `self_check()` tests, is:

$$\begin{aligned}
 \phi &\leftarrow \text{backbone}(x). \\
 \mu_k &\leftarrow m_k^\top \phi, \quad v_k \leftarrow \phi^\top \text{diag}(v_k) \phi + \sigma_k^2. \\
 \text{bound} &\leftarrow \mu_y - \log \sum_k \exp(\mu_k + \tfrac{1}{2} v_k). \\
 \text{loss} &\leftarrow -\text{mean}(\text{bound}) + (1/T) \text{KL}(q||p) - (1/T) \log p(\sigma^2).
 \end{aligned}$$

Before training, `self_check()` builds an 8×4 random batch and checks three facts: the loss is finite and back-propagates; the bound matches a hand-expanded log-sum-exp; multiplying the features by

four increases logit variance. That check is the implementation test. It does not use Fashion-MNIST accuracy as a substitute for the formula.

Practical settings follow the paper and the authors’ diagonal parameterization rather than a new variant. The prior variance is $2/D$. Weight standard deviations are initialized near $1/D$. The KL is scaled by $1/T$ in the main runs. I did not import the `vbll` library; the module is a direct implementation of the bound above so the loss can be read against Equation (2).

4 Experiment Settings and Results

Task. In-distribution data are Fashion-MNIST clothing images, 28×28 greyscale, 10 classes [2]. The official training set is split once, with seed 0, into 54,000 training images and 6,000 validation images. The 10,000 official test images are used only for the tables below. MNIST digits are a far out-of-distribution set [3]: same resolution, different pattern family. Both datasets are normalized with the Fashion-MNIST training mean 0.286 and standard deviation 0.353.

Protocol. The three heads share the CNN, the augmentation (horizontal flip and a random crop after 2-pixel padding), AdamW at learning rate 10^{-3} , cosine decay, batch size 128, and 12 epochs. Backbone weight decay is 10^{-4} . MAP and D-VBLL use seeds $\{0, 1, 2\}$. G-VBLL uses seeds $\{0, 1\}$. All runs are on Apple MPS (PyTorch 2.14). Validation probabilities for D-VBLL use 16 logit samples; the test tables use 32. MAP and D-VBLL means are over three seeds. G-VBLL means are over two. Calibration is expected calibration error with 15 bins [5].

Table 2: Clean Fashion-MNIST test set. MAP and D-VBLL are means over three seeds; G-VBLL over two.

Model	Accuracy	NLL	ECE	Entropy
Softmax MAP	0.931	0.199	0.0104	0.167
D-VBLL ($1/T$)	0.928	0.201	0.0074	0.183
G-VBLL	0.928	0.206	0.0131	0.252

On clean clothes the three heads are tied on accuracy. D-VBLL’s ECE stays near 0.007 on every seed, below MAP. G-VBLL is slightly worse on NLL and ECE and has higher entropy, which is the price of the wider class-conditional Gaussians. The absolute gap is small because the MAP model is already well calibrated. Figure 2 shows the shared learning curve. Figure 3 shows that D-VBLL’s reliability curve sits closer to the diagonal through the middle bins.

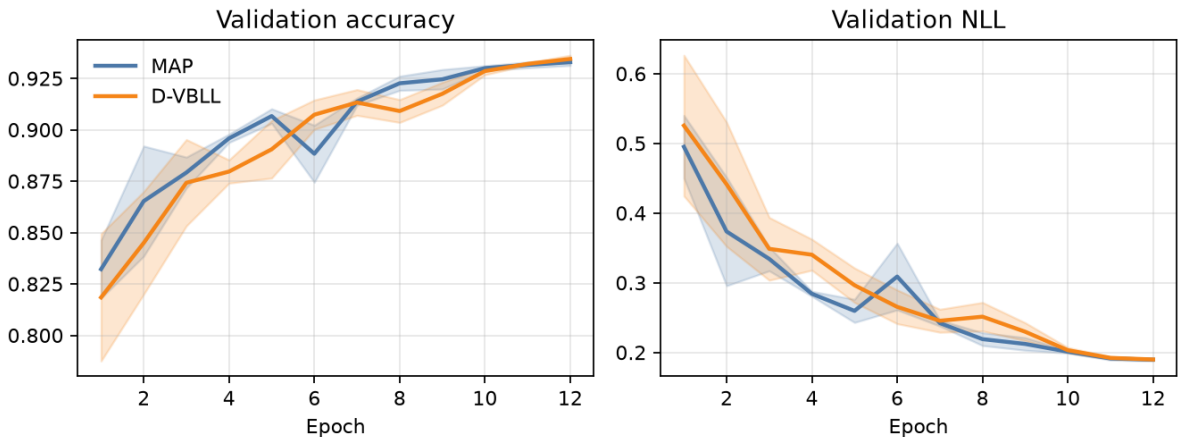


Figure 2: Validation accuracy and NLL for MAP and D-VBLL. Bands are three seeds. The training losses are not plotted together because cross-entropy and the ELBO are different objectives.

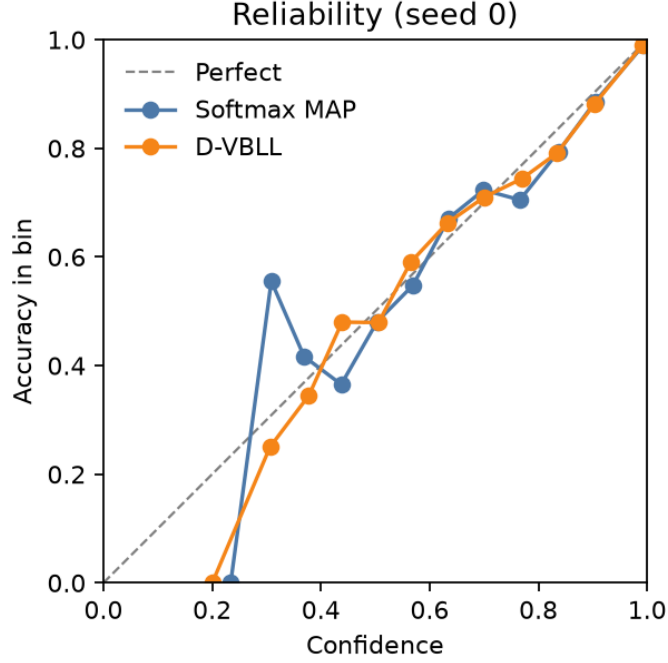


Figure 3: Reliability diagram on the clean test set, seed 0. Both models are close to the diagonal; D-VBLL is the smoother of the two.

Controls a probability model still has to pass. A lower ECE than softmax is not, by itself, evidence that the posterior did the work. Three controls use the same test set and do not change the trained CNN.

Temperature scaling [5] divides the MAP logits by a scalar T fit on the 6,000 validation images by minimizing NLL. It does not change the predicted class. On the three MAP seeds, T is 1.114, 1.088, and 1.119. Test ECE falls to a mean of 0.0054, below D-VBLL’s 0.0074, and mean NLL is 0.197. The ECE gap in Table 2 is therefore mostly a temperature, which matches the later finding that $q(W)$ has concentrated.

Gaussian naive Bayes on raw pixels, with an independent Gaussian per pixel and per class, reaches accuracy 0.579, NLL 7.63, and ECE 0.419. The independence assumption does not describe a garment. Shrinkage linear discriminant analysis on the 128-d features of the seed-0 MAP network is the classical shared-covariance Bayes classifier on the same representation. It reaches accuracy 0.917, close to the discriminative heads, but NLL 0.544 and ECE 0.063. The CNN features carry the class. The softmax and D-VBLL heads are what make the probabilities sharp.

Ranking is a separate question from calibration. Using confidence to detect the model’s own mistakes, mean AUROC is 0.922 for MAP, 0.924 for D-VBLL, and 0.922 for G-VBLL. Pixel naive Bayes is at chance on this ranking (0.510); LDA on the CNN features reaches 0.900. Keeping the most confident 90% of the clean test set raises accuracy to 0.968 (MAP), 0.967 (D-VBLL), and 0.967 (G-VBLL). Figure 4 shows the risk-coverage curves lying on top of each other. D-VBLL does not rank errors better than softmax once the posterior has collapsed. Both heads are still useful as reject options: the bottom decile of confidence contains a disproportionate share of the mistakes.

Table 3: Clean test controls. Neural rows are means over the runs in Table 2. Temperature is fit once per MAP seed. Naive Bayes and LDA are single fits. Brier is the mean squared error of the probability vector.

Model	Accuracy	NLL	ECE	Brier	Error AUROC
Gaussian naive Bayes (pixels)	0.579	7.630	0.419	0.840	0.510
LDA on MAP features	0.917	0.544	0.063	0.143	0.900
Softmax MAP	0.931	0.199	0.0104	0.103	0.922
MAP + temperature	0.931	0.197	0.0054	0.103	0.922
D-VBLL (1/T)	0.928	0.201	0.0074	0.105	0.924
G-VBLL	0.928	0.206	0.0131	0.107	0.922

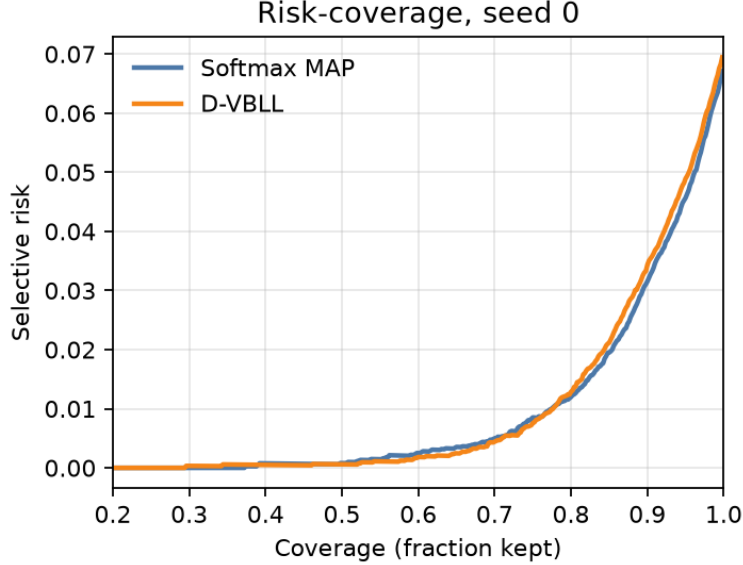


Figure 4: Selective risk against coverage, seed 0. Lower is better. The curves overlap: both heads become accurate once low-confidence predictions are refused.

Where the errors are. Seed-0 D-VBLL per-class accuracy is high on bags (0.989), trousers (0.987), and sandals (0.978), and lowest on shirts (0.769). The largest off-diagonal counts are shirt→T-shirt (96), T-shirt→shirt (79), shirt→coat (70), and coat→shirt (41). Those pairs share a torso silhouette. Ankle boot→sneaker (40) is the main footwear confusion. Figure 5 is the row-normalized matrix. Figure 6 shows the failure mode the ECE number hides: a few errors still receive predictive probability near 1, including a boot read as a sneaker.

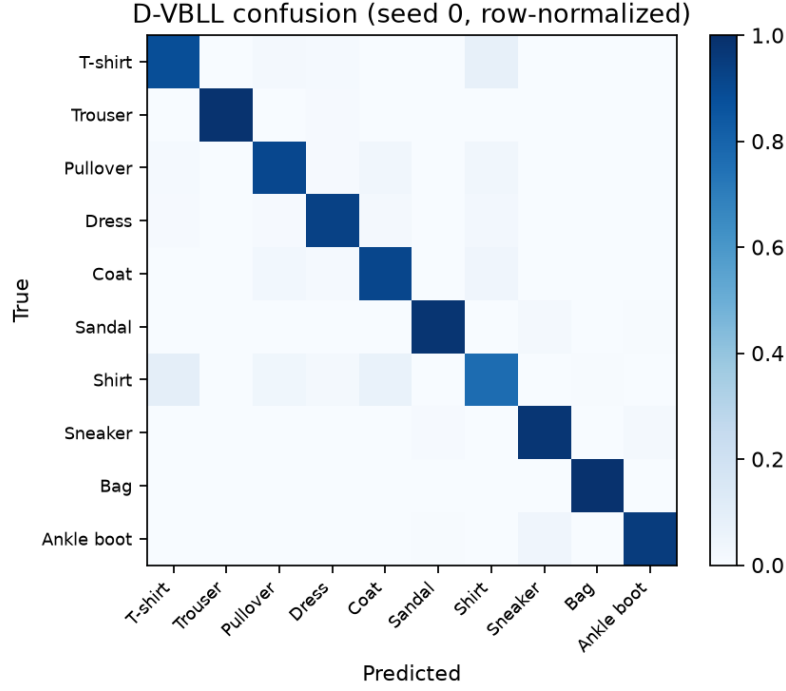


Figure 5: D-VBLL confusion on the clean test set, seed 0. Mass off the diagonal sits in the shirt / T-shirt / coat / pullover block.

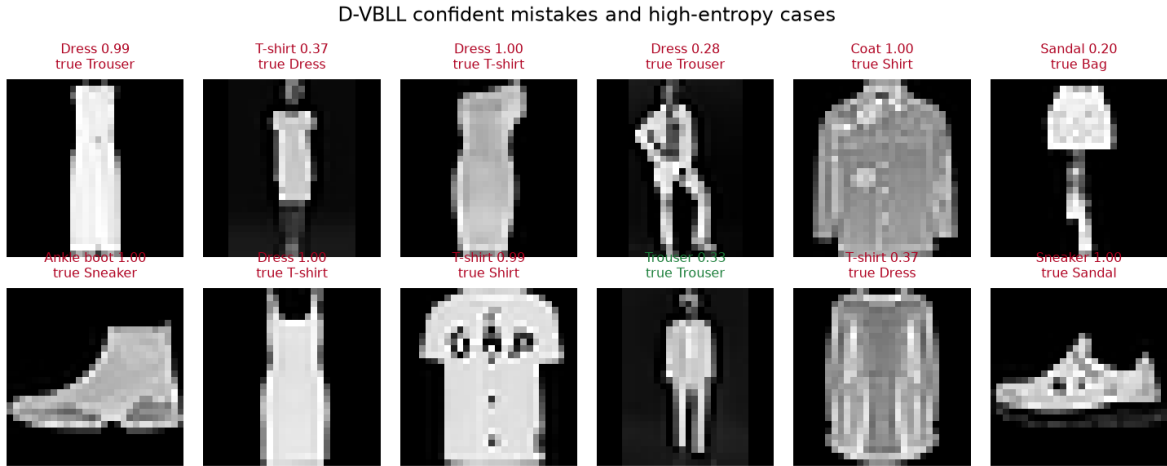
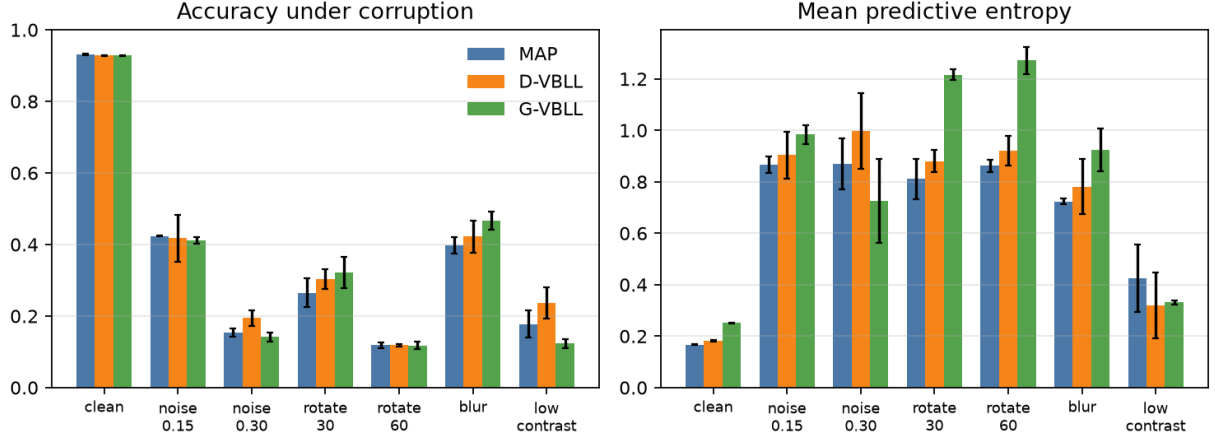


Figure 6: D-VBLL seed 0. Titles are the predictive class, its probability, and the true class. Red titles are mistakes, including high-confidence ones.

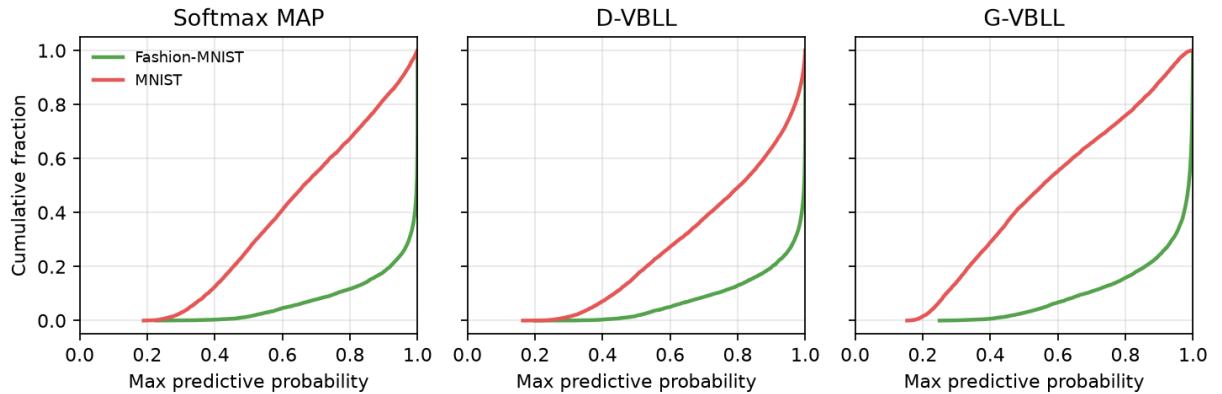
Corruptions. Corruptions are applied in pixel space at test time only. The models were not trained on them. Table 4 and Figure 7 show that geometric and photometric shift hurts all three heads. At noise 0.30, rotation 60° , and low contrast, accuracy falls to roughly 0.12–0.23. Low contrast is the instructive case: accuracy is about 0.19–0.23, but mean entropy stays near 0.33, lower than under noise, and ECE exceeds 0.65. All three heads are confidently wrong when the silhouette is preserved and the intensity is flattened. Last-layer weight uncertainty is not a corruption detector.

Table 4: Test accuracy under corruption. MAP and D-VBLL are means over three seeds; G-VBLL over two.

Condition	MAP	D-VBLL	G-VBLL
Clean	0.931	0.928	0.928
Gaussian noise 0.15	0.423	0.418	0.412
Gaussian noise 0.30	0.154	0.195	0.141
Rotate 30°	0.264	0.303	0.322
Rotate 60°	0.118	0.118	0.118
5 × 5 average blur	0.397	0.423	0.466
Low contrast	0.178	0.237	0.123

**Figure 7:** Accuracy falls under every corruption for all three heads. Entropy rises under noise and rotation, but stays low under low contrast.

Out-of-distribution digits. Using maximum predictive probability as the in-distribution score, mean AUROC against MNIST is 0.851 for MAP (standard deviation 0.039 across three seeds) and 0.856 for D-VBLL (standard deviation 0.026). The third seed removes the apparent gap between those two heads. G-VBLL reaches 0.910 on both of its seeds (standard deviation 0.003). Figure 8 is a cumulative plot so the mass near probability 1 does not hide the rest of the distribution. G-VBLL’s MNIST curve rises earlier: more digits receive a lower maximum probability, which is what the AUROC is measuring.

**Figure 8:** Cumulative distribution of maximum predictive probability, seed 0. A curve that rises earlier is less confident. G-VBLL separates MNIST from Fashion-MNIST more cleanly.

Why the Bayesian head stays close to MAP. After training at KL weight $1/T$, the mean diagonal variance of W is about 4.9×10^{-4} . The prior variance is $2/D = 0.0156$, roughly thirty times larger, so the posterior has concentrated. The KL itself is about 2.3×10^3 ; divided by $T = 54000$

it contributes only a few hundredths to the loss, which is enough to regularize but not enough to keep the prior’s uncertainty. Figure 9 then has a simple shape for both models: predictive entropy falls as $\|\phi\|$ grows. Large features produce large mean logits, and the quadratic variance term is too small to cancel that peak. This is the opposite of a distance-aware uncertainty method, and it matches the modest ECE gap in Table 2.

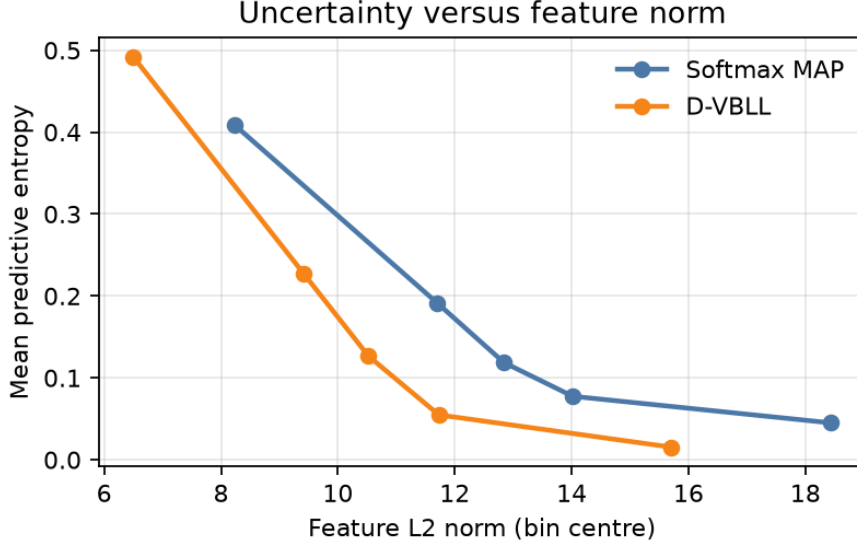


Figure 9: Clean-test entropy against feature norm. Higher-norm features are more confident for both heads.

KL-weight ablation. The $1/T$ factor is a hyperparameter of the ELBO estimator, not a constant of nature. D-VBLL was retrained at $10/T$ and $100/T$ for seeds 0 and 1. Table 5 reports means. A single seed had made $10/T$ look like an OOD improvement (0.869 versus 0.825). The second seed removes that claim: mean AUROC at $10/T$ is 0.844, which does not beat $1/T$ (0.856). $100/T$ does raise mean AUROC to 0.913, and it also raises ECE to 0.094 and NLL to 0.288. Accuracy stays at 0.927–0.928. The KL term is alive. A moderate increase does not buy out-of-distribution detection, and a large increase buys it by making the predictive distribution less sharp on the clean test set.

Table 5: D-VBLL KL multiplier. $1/T$ is the mean of three seeds. $10/T$ and $100/T$ are means of seeds 0 and 1.

KL weight	Accuracy	NLL	ECE	Entropy	MNIST AUROC
$1/T$	0.928	0.201	0.0074	0.183	0.856
$10/T$	0.928	0.211	0.0160	0.277	0.844
$100/T$	0.927	0.288	0.0938	0.637	0.913

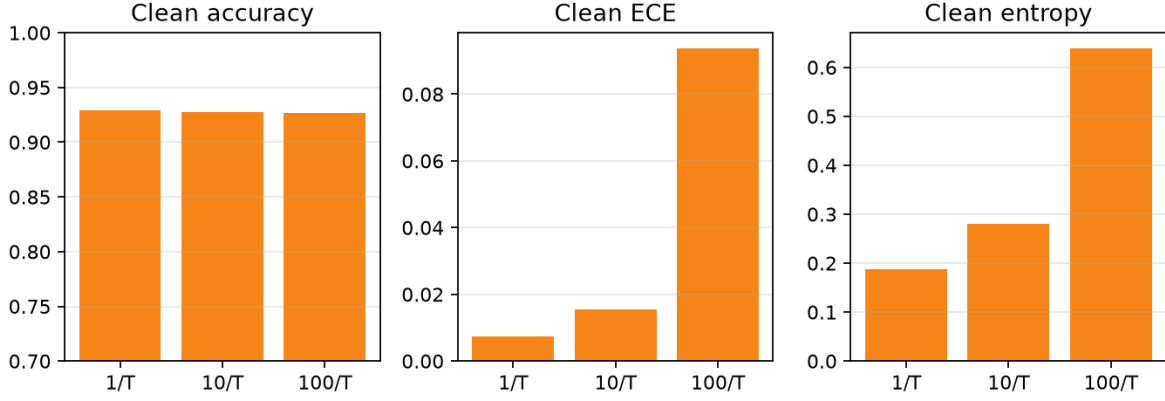


Figure 10: Seed-0 ablation of the KL multiplier. $1/T$ is the main setting. Table 5 is the multi-seed version of the same sweep.

Prior scale does not uncollapse the posterior. The prior variance is $\text{prior_scale} \times 2/D$. I retrained seed-0 D-VBLL at scales 0.1, 1, and 10, so the prior variance runs from 0.0016 to 0.156. Mean posterior variance stays between 2.5×10^{-4} and 5.4×10^{-4} in all three runs. Widening the prior does not restore the uncertainty the KL term at $1/T$ is too weak to protect. Entropy and MNIST AUROC both fall as the prior widens: AUROC is 0.883, 0.825, and 0.791. The KL multiplier, not the prior scale, is the knob that moved the predictive distribution.

Table 6: D-VBLL prior scale, seed 0, KL weight $1/T$.

Scale	Accuracy	NLL	ECE	Entropy	MNIST AUROC
0.1	0.928	0.204	0.0064	0.219	0.883
1	0.929	0.201	0.0075	0.187	0.825
10	0.930	0.203	0.0089	0.178	0.791

Monte Carlo sample size. D-VBLL predictions average softmaxes. On the three seeds, 8, 32, and 64 samples agree to about 0.001 in accuracy and NLL, and ECE stays inside 0.007–0.008. The published tables use 32 samples. Eight would have been enough on this dataset.

Shirt versus T-shirt. Headline ECE near 0.01 is an average over easy classes. On shirt images alone, accuracy is 0.776 (MAP), 0.767 (D-VBLL), and 0.760 (G-VBLL), and the class-conditional ECE is about 0.074, 0.075, and 0.049. About 10% of shirts are predicted as T-shirts under every head. G-VBLL is less confident on shirts (mean confidence 0.79 against 0.84 for MAP), which is why its shirt ECE is lower, not because it names them more often. The overlap is in the labels and the silhouette.

Rejection under shift. Dropping the least confident 10% helps on clean images and does not help once the input is shifted. At 60° rotation, accuracy among the kept 90% is 0.122–0.123 for all three heads, against a full-set accuracy of 0.118. Under low contrast the kept slice is 0.174 (MAP), 0.241 (D-VBLL), and 0.119 (G-VBLL), against full-set accuracies of 0.178, 0.237, and 0.123. Figure 11 is that comparison. Confidence is a reject option when the test distribution matches training. It is not a detector for a preserved silhouette with the wrong intensity or the wrong angle.

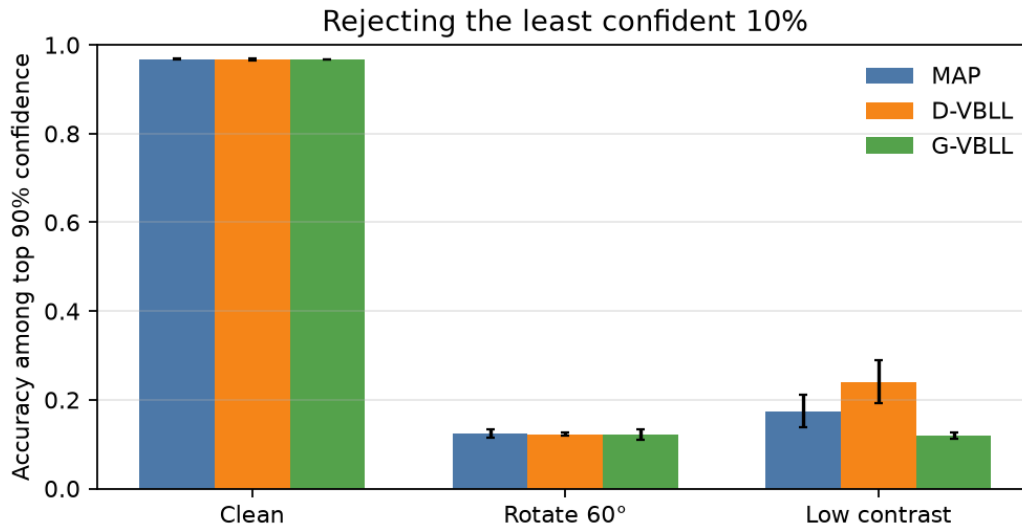


Figure 11: Accuracy after keeping the most confident 90%. The gain exists on clean clothes and disappears under 60° rotation and low contrast.

5 What I Have Learnt from This AI Assignment

- The useful constraint was the brief itself: probability or Bayes, recent, and not a repeat of FasterNet. Once that was stated, the agent could reject a fast CNN and look for a last-layer posterior with a checkable formula.
- A Bayesian classifier is defined by the predictive distribution, not by the word “uncertainty” in the README. Here that distribution is an average of softmaxes under $q(W)$. The self-check on Equation (2) mattered more than the first epoch’s accuracy.
- Matching accuracy is a real result, and it is not the whole probability question. D-VBLL at $1/T$ ties MAP on accuracy, NLL, Brier score, and error ranking. The ECE edge over unscaled softmax disappears once the softmax has a validation temperature. I should not describe D-VBLL as a better calibrated classifier than temperature scaling.
- Posterior variance can collapse, and widening the prior does not undo it. The prior said 0.016 at the default scale; training at $1/T$ kept the posterior variance below 6×10^{-4} even when that prior was widened or narrowed by a factor of ten. A second seed also removed a false OOD win for the $10/T$ KL weight. Only $100/T$ raises MNIST AUROC, and it does so by making clean probabilities less sharp.
- A Bayes classifier still needs a representation. Pixel naive Bayes scores 0.579 because garment pixels are not independent Gaussians. LDA on the CNN features jumps to 0.917, and the discriminative heads then cut NLL from 0.544 to 0.200. The modern part is the feature map; the last-layer posterior is a small adjustment on top.
- The generative head is the one that uses Bayes’ rule, and it is the one that moves out-of-distribution detection. G-VBLL’s MNIST AUROC is 0.910 with almost no seed spread. It pays for that with a higher clean NLL and ECE than D-VBLL. LDA on the same style of model, shared covariance on frozen features, reaches accuracy 0.917 and then loses on NLL. The formula and the representation both matter.
- Confidence is a reject option only on the training distribution. Dropping the least confident 10% lifts clean accuracy to about 0.967 and does nothing under 60° rotation or low contrast. Shirt

ECE stays near 0.07 while the headline ECE is near 0.01. Eight Monte Carlo samples match sixty-four on this test set.

- Out-of-distribution AUROC needs more than one seed. MAP’s three seeds still spread by about nine points. G-VBLL’s two seeds differ by 0.003. A single-run bar chart would have picked a winner that the next seed does not support.

6 Source Code Webpage Link

Source code:

<https://github.com/AngusJai/CISC3024-AI-Assignment2>

The repository contains the model code, the training and evaluation scripts, `process_log.md`, `report.tex`, and the metrics and figures under `outputs/`. Dataset downloads and virtual environments are gitignored. Trained checkpoints are included so `demo.py` can be run without retraining.

7 Conclusion

D-VBLL is a recent Bayesian classifier that an agent can implement from a variational bound and test on clothing images in one afternoon. With the paper’s $1/T$ KL weight it reproduces the softmax baseline’s accuracy and slightly improves unscaled calibration. A validation temperature on that softmax calibrates further, and a classical LDA head on the same features is close in accuracy but worse in NLL. D-VBLL does not detect digit images, because the approximate posterior concentrates. Widening the prior leaves that concentration in place. Only a much larger KL weight raises MNIST AUROC, and it does so by hurting clean NLL and ECE. G-VBLL, the generative classifier in the same paper, is the head that applies Bayes’ rule to the features. Its MNIST AUROC is 0.910 and stable across seeds, with a small cost in clean likelihood and calibration. Refusing the least confident tenth of the test set helps only when the images still look like the training set. Accuracy on this network stays near 0.93.

References

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